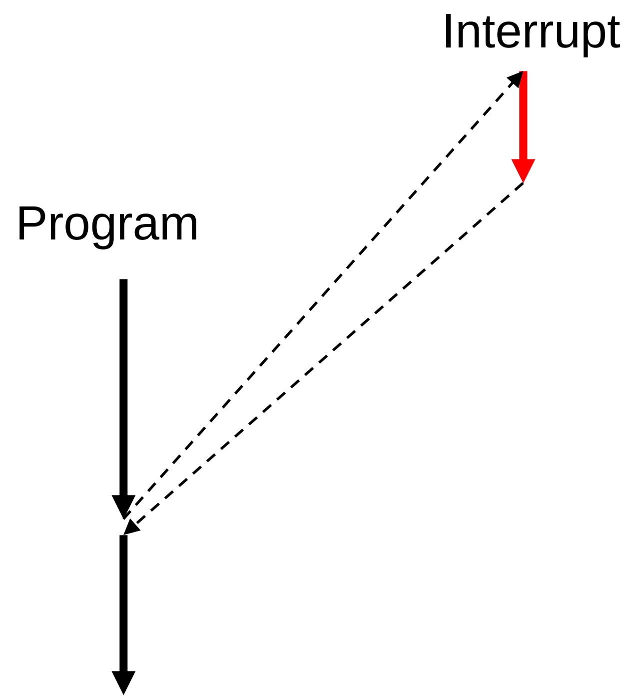
Interrupts

- Interrupts are a mechanism which makes processor to respond quickly to external (or internal) stimulus
  - For example when IO-device switches to ready-state
- Current execution is interrupted and processor jumps to interrupt handler code. After handler is executed processor resumes execution at the address where interrupt occured
- Interrupt is ”invisible”. The program that is interrupted is not interfered by interrupt handler (except for the delay caused by interrupt handler execution)



Interrupt is hardware coerced subroutine

- Ordinary subroutine needs only to save registers that calling convention defines as callee saved
  - C/C++ compiler takes care of saving registers for you
- When you call an ordinary subroutine you can prepare for subroutine call by saving important registers that may be modified by the subroutine
- Interrupts are triggered independently of the program execution thus interrupt handler needs to ensure that interrupted program is not disturbed
  - Interrupt handler needs to save **all** registers that it modifies
  - **Flags** must be preserved
    - For example interrupt can occur between comparison instruction and conditional branch
  - When an interrupt routine is written in C/C++ you must tell the compiler that the routine is an interrupt handler

    GCC: `void handler() __attribute__ ((interrupt ("IRQ")));`
  - Our ARM cortex-M3 is an exception (see next slide)

ARM Cortex-M3 interrupt handling

- ARM Cortex-M processors target low-end and mid-range embedded systems
- ARM has implemented some extra hardware measures to simplify the way interrupt handlers are written
  - ARM has laid out recommened register usage convention that defines which registers need to be saved by a called function
  - ARM has implemented hardware that saves the other registers automatically before interrupt handler is entered
  - If compiler follows ARM register usage convention interrupt handler can be an ordinary function (no special attributes are needed)
    - GCC follows ARM register usage convetion

AAPCS register usage

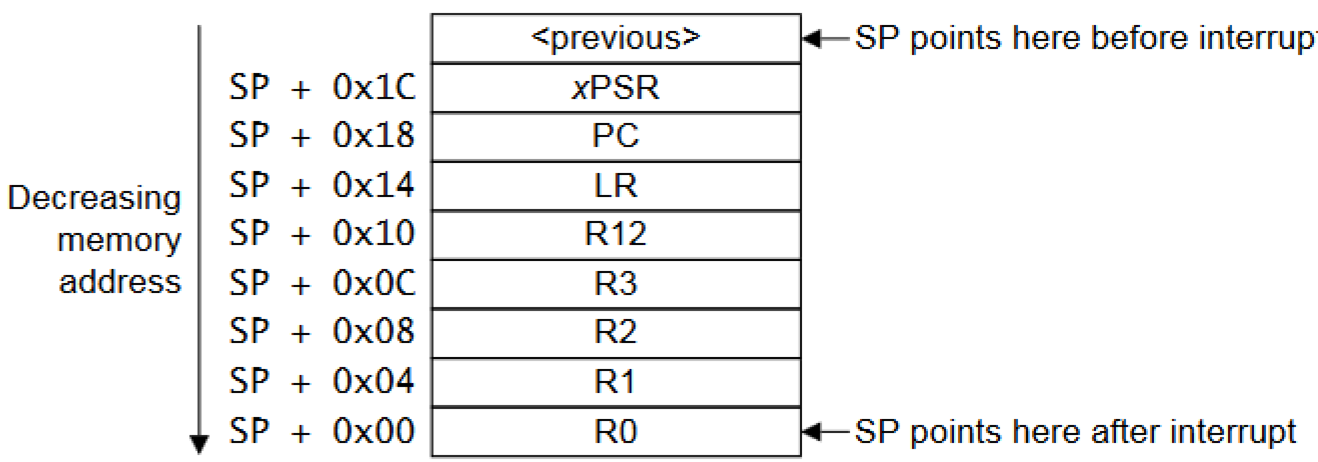
- ARM standard requires that when a function is called the function must preserve the contents of registers R4 – R11.
  - A function written in C/C++ will save and restore them if they are used by the function

AAPCS compliant compiler will preserve these registers

Register	Synonym	Special	Role in the procedure call standard
r15		PC	The Program Counter.
r14		LR	The Link Register.
r13		SP	The Stack Pointer.
r12		IP	The Intra-Procedure-call scratch register.
r11	v8		Variable-register 8.
r10	v7		Variable-register 7.
r9		v6 SB TR	Platform register. The meaning of this register is defined by the platform standard.
r8	v5		Variable-register 5.
r7	v4		Variable register 4.
r6	v3		Variable register 3.
r5	v2		Variable register 2.
r4	v1		Variable register 1.
r3	a4		Argument / scratch register 4.
r2	a3		Argument / scratch register 3.
r1	a2		Argument / result / scratch register 2.
r0	a1		Argument / result / scratch register 1.

Automatic register stacking

- When an interrupt occurs the processor automatically pushes some of the register on the stack
  - Since compiler will preserve R4 – R11 they don’t need to be automatically saved
  - Only the registers that compiler is free to use are automatically pushed on the stack

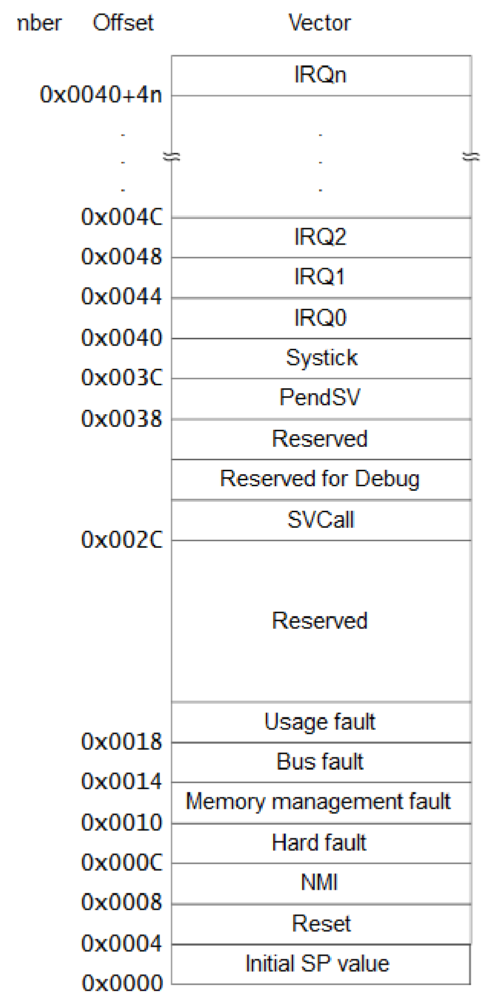


Interrupt handler

- Processors have multiple interrupt sources
- Interrupt handler needs to find out which of the sources caused the interrupt and then handle the interrupt
  - Handling means that interrupt is acknowledged and the action associated with interrupt is performed (for example data transfer)
  - Interrupt is acknowledged by
    - Bus activity
      - Read or write to a special address (for example status register)
    - External signal
    - Some sources need no acknowledge at all. For example systick timer needs no acknowledge
- Most modern microcontrollers use interrupt vectors
  - Each type of interrupt has a dedicated memory location where the interrupt handler address is stored. Hardware automatically calls the appropriate handler.

Cortex-M3 interrupt vectors

- Interrupt vector is the address to which hardware jumps when an interrupt occurs
- Compiler/linker fills in the vector table with addresses of the interrupt handlers
    - PicoSDK uses a system where vectors are identified with a vector number. PicoSDK provides API for managing user interrupts using defines instead of raw numbers and provides mechanism for shared interrupt handlers.
    - User interrupts start from IRQ0 which in RP2040 is TIMER_IRQ_0 (see hardware_irq/irq.h)
- It is also possible to set non-shared handlers



Interrupt handler

- Save processor state
  - Return address (some processors do this automatically)
  - General purpose registers
  - Flags
  - Special registers (processor dependent)
  - Stack pointer

Handled by hardware/compiler/PicoSDK

- Acknowledge interrupt
- Execute required action (read/write data)

- Restore processor state
- Return to location where interrupt occured

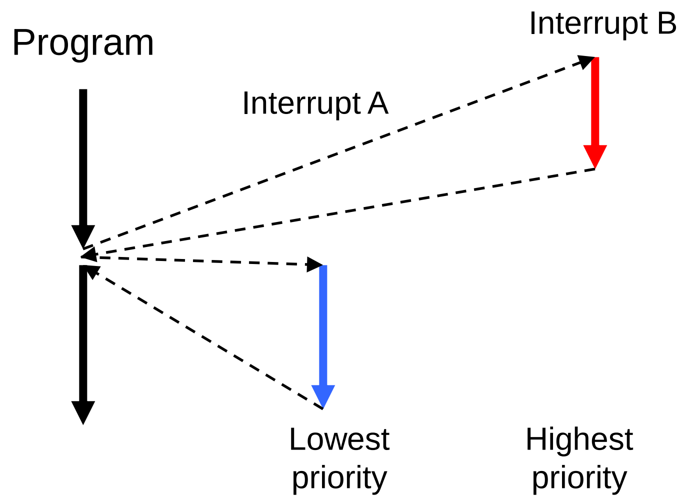
Handled by hardware/compiler/PicoSDK

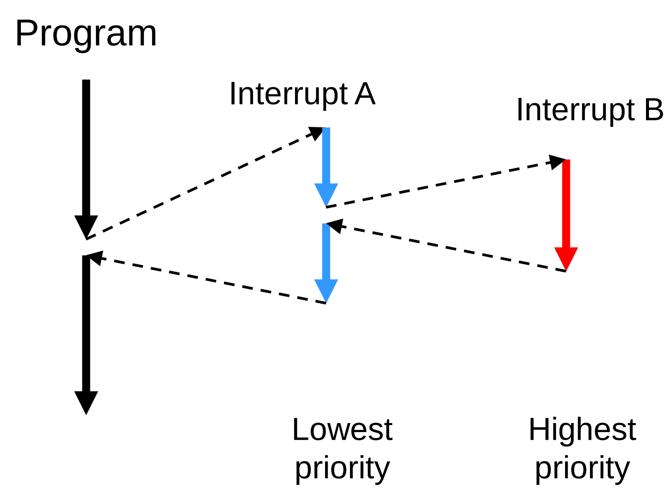
Interrupt categories

- Device interrupts (hardware interrupts)
  - Occurs asynchronously
  - From external or integrated peripheral device (UART, AD-converter, NIC, etc.)
- Interrupts caused by software
  - Current instruction causes interrupt
    - Software interrupt (explicit request)
    - Others:
      - Illegal instruction
      - Illegal memory access
      - Page fault
      - Etc.

Interrupt priorities

- Interrupts have priorities which determine the order of handlers if two interrupts occur simultaneously
  - Higher priority is executed before lower priority
- Interrupts can also be nested
  - Higher priority interrupt can interrupt lower priority handler
    - Requires larger stack
- Priorities are assigned by programmer or hardware designer
  - Some processors have fixed priorities that can not be changed by the software





Interrupts can be disabled

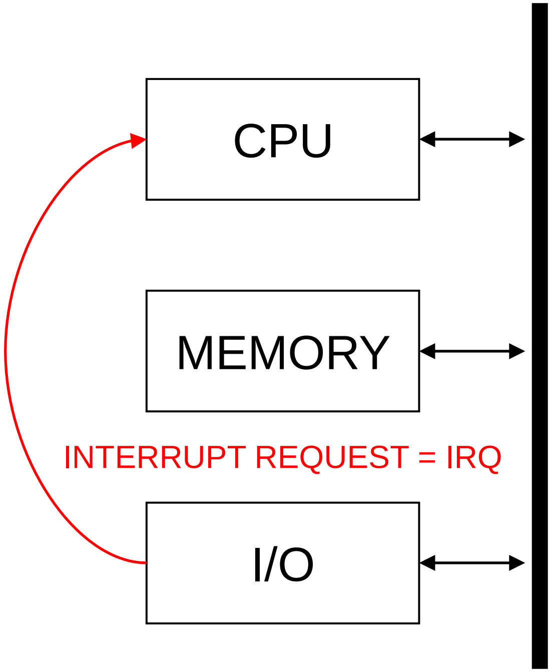
- Some parts of program may not be interrupted
  - Such a part is called a critical section
  - Interrupts must be disabled during execution of critical section
    - Increases interrupt response time = time between interrupt signal and handler execution
    - Minimize the time that interrupts are disabled – make your critical section as short as possible

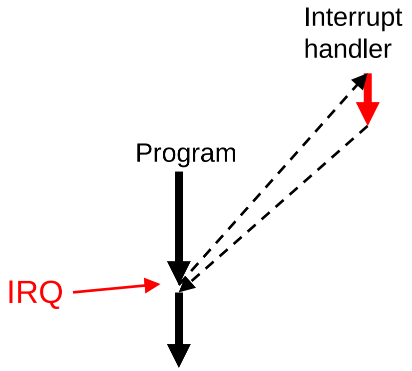
Interrupts and privileges

- If a processor has at least two privilege levels (user mode and privileged mode) the interrupts handler runs in the privileged mode
  - User mode access to hardware resources is restricted. For example processor may prevent access to IO-devices, memory management and prevent user from changing privilege level
  - In privileged mode there are no access restrictions
  - Operating system kernels run in privileged mode
    - Whether user programs run in privileged mode depends on the operating system. Unis style operating systems restrict user privileges while may RTOSes allow programs to run in privileged mode as well
    - Operating system calls can be implemented with software interrupts which switches to privileged mode automatically on OS calls

Interrupt driven IO

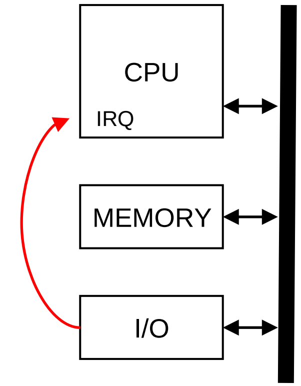
- IO device is connected to the interrupt input of the CPU
- When device needs attention it asserts the interrupt signal
- Processor interrupts the currently exectuing program and jumps to interrupt handler
- Interrupt handler executes the actions that the device requires and acknowledges the interrupt
  - For example copies the data from IO device
  - If further processing is needed that is done in the main program – the handler and main program need to exchange some information

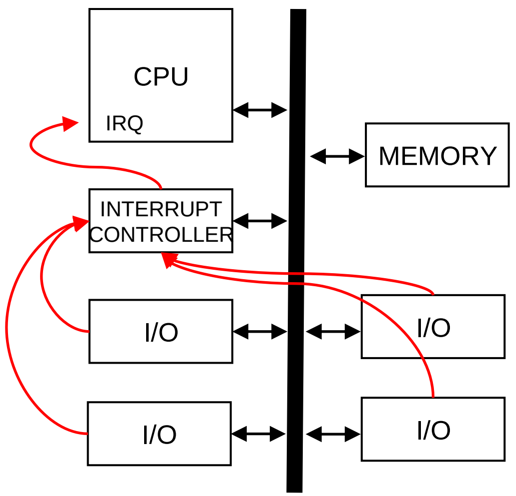




Interrupt driven IO

- Interrupt controller receives interrupt requests and passes them to the CPU
  - Can handle several different types of interrupt signaling
  - Prioritizes the interrupts
  - Keeps track of pending interrupts
  - Has storage for interrupt vectors (handler functions addresses)
  - In a modern microcontroller the interrupt controller is integrated to the CPU

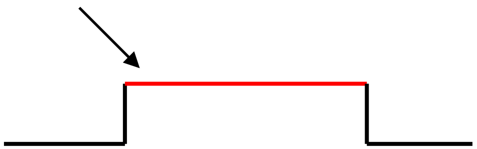




Interrupt signal types

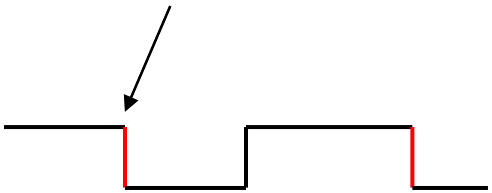
- Level sensitive interrupt
  - Signal level indicates interrupt request
  - Request is active as long as signal is active
  - Interrupt handler must acknowledge interrupt to deactivate interrupt request
- Edge sensitive interrupt
  - Change of signal indicates interrupt request
- Type of interrupt signal is configured by the programmer
  - Can (typically) be configured for each interrupt source
    - For external signals

Interrupt is active as long as signal is high



Active high request

Signal must go high before new interrupt can occur



Falling edge triggered interrupt

Interrupt driven IO

- USB keyboard is attached to a microcontroller that runs at 72 MHz
  - USB HID transfer rate is max 64 kbps (8 kilobytes per second)
  - One interrupt takes 300 clock cycles and transfers one character
  - If user types twelve eigth character words per minute (~100 chars per minute) then the interrupt is executed only 100 times

$$\frac{100 \times 300}{60 \times 72 \times 10^6} \approx 0{,}0000694\,\%$$

- Benefits of interrupt driven IO
  - Fast response time
  - Processor resources are used only when needed

How to communicate with an interrupt handler

- If an operating system is used then OS primitives are used for communication
  - A typical RTOS provides signals and queues for communication
- If there is no operating system then a shared memory region (variables, structure or object)
  - The simplest way is to set a flag (= global variable) in the interrupt handler and when the main program sees the set flag it does its part of the processing
    - Can cause race condition
    - Makes sense only in a small system where timings can be fully analyzed
  - Usually ISR puts data/events in a queue
    - For example received characters go into a buffer where they can be retrieved and processed by the main program
  - An operating system uses similar mechanisms but hides them behind the OS API

Critical section

- Access to a shared memory region must be an atomic operation
  - While main program is accessing the shared memory ISR is not allowed to change any values →critical section
- Think of a case where ISR and main program share a buffer for data and a counter to indicate the amount of data in the buffer

Main program:

```
if(counter > 0) {
  data = get_value_from_buffer();
  counter = counter - 1;
}
```

3 1

Incrementing/decrementing a variable is a **read-modify-write** sequence which must be atomic!

What happens if ISR executes after (1) but before the assigment?

ISR (interrupt handler):

2

```
data = read_from_io_device();
put_value_in_buffer(data);
counter = counter + 1;
```

Critical section

- On a single core processor a critical section can be implemented by
  - disabling interrupts when entering critical section and
  - enabling/restoring the interrupt state when leaving the critical section.

Priciple of using critical section on a single core

```
// returns the interrupt enable state before entering critical section
bool enter_critical(void);
// restore interrupt enable state
void leave_critical(bool enable);
```

```
bool irq;

irq = enter_critical();

if(counter > 0) {
    data = get_value_from_buffer();
    counter = counter - 1;
}

leave_critical(irq);
```

Interrupts are disabled while this is executed.

- If you use critical sections keep them as short as possible
- Disabling interrupts increases interrupt latency (reaction time)

Sometimes critical section can be avoided

- In many cases the use of critical sections can be avoided all together
- A classic example is a ring buffer with head and tail pointers
  - Producer modifies only head pointer and reads tail pointer
  - Consumer modifies only tail pointer and reads head pointer
  - This way there is only one writer for each pointer which means that writes can never be corrupted by interrupts
- Multi-core atomic memory access is a completely different story
  - There are means for that in some ARM architectures but that is out of the scope of this course
  - RP2040 is a multicore processor with M0+ cores
    - The cores don’t have support for multicore critical sections
    - Critical sections are implemented in the SDK using combination of external spin-lock peripheral and disabling interrupts
  - Critical section should be as short as possible

Interrupt latency

- Interrupt latency has two components: the execution time of higher priority interrupts and the time that interrupts are disabled
- How to estimate the worst case latency?
  - Assume that all higher priority interrupts are triggered at the same time → sum up their execution times
  - Add the longest time that interrupts are disabled anywhere in the program (a single case, not a sum of disable times)
  - Analyze if your ISR can starve – if higher priority interrupts are triggered at such a high rate that the ISR in question never gets to execute you are in big trouble
- In most cases the worst case will not occur very often – but in a time critical system you must be prepared for it

Make ISR as quick as possible

- The ISR should be as short as possible and do only time critical things that can't be delayed to a later time
- Three things that ISR must avoid
  - Wait for something – an ISR that needs to wait indicates incomplete analysis of system behaviour
  - Allocate or free memory (new/delete or malloc/free)
  - Call library functions (print, file handling etc.)
- The golden rule of ISR programming: Get in – get out (quickly)

```
void pulse_isr(void)
{
  volatile int channels;

  /* read which channel(s) caused interrupt */
  channels = LPC21_T0IR;
  /* clear interrupt from timer HW */
  LPC21_T0IR = channels;

  pulses++;

  if(--precounter == 0) {
    precounter = prescaler;
    old_length = pulse_length;
    pulse_length = LPC21_T0CR0;
  }
}
```

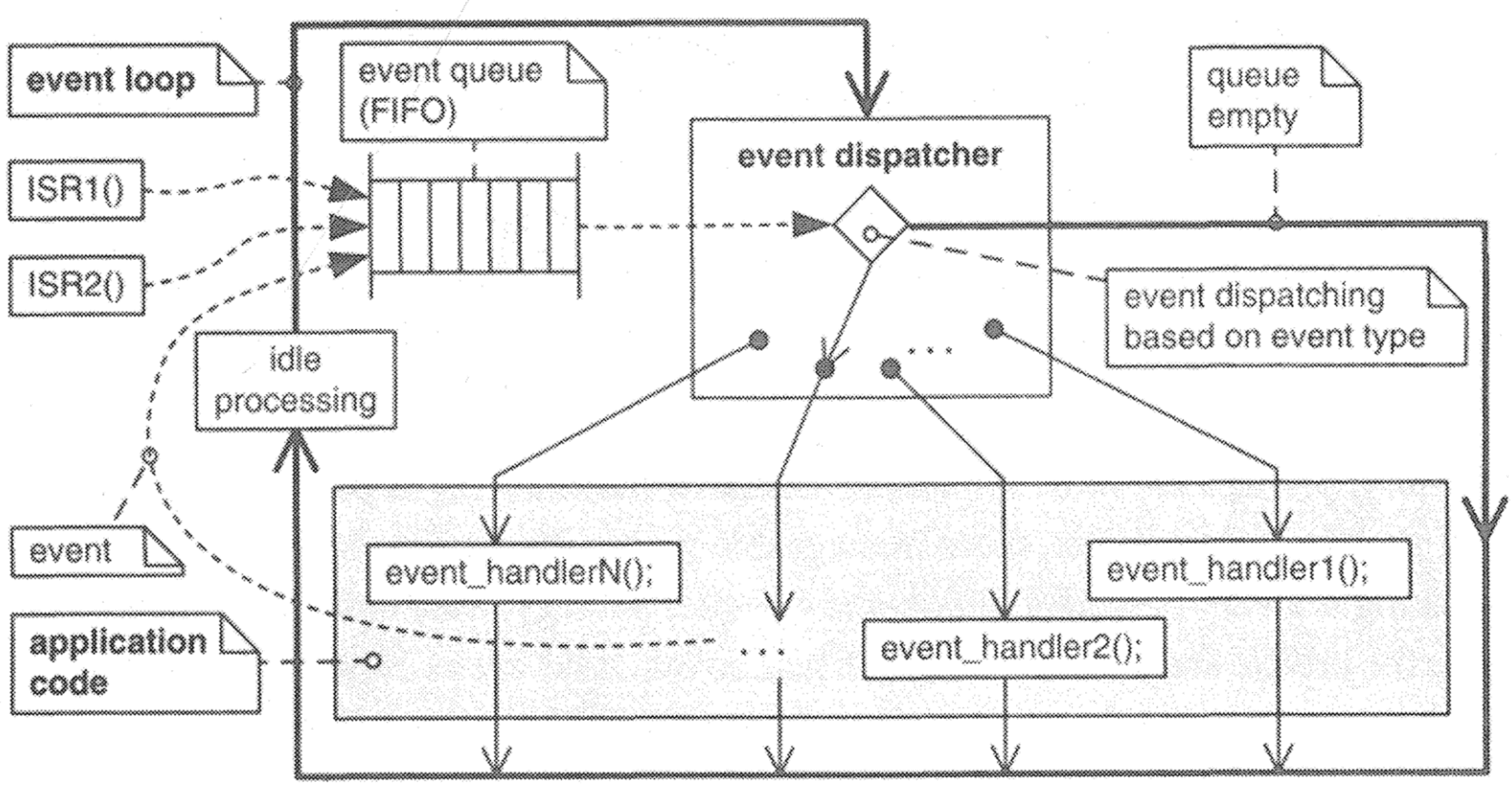
```
void uart_handler(int uart_nr)
{
    int interrupt;
    int status;
    UART_PARAMS *u;
    UART_STATE *s;

    u = &uart[uart_nr];
    s = &state[uart_nr];

    do {
        interrupt = *u->iir;
        if ((interrupt & 0x01) == 0) {
            status = *u->lsr;
            if (interrupt & INTR_XMT_EMPT) {
                if (status & LSR_THRE) {
                    uart_transmit(u, s);
                }
            }
            else {
                while (status & LSR_RECEIVE) {
                    if (status & LSR_RECEIVE_ERROR) {
                        uart_error(u, s);
                    }
                    else {
                        uart_receive(u, s);
                    }
                    status = *u->lsr;
                }
            }
        }
    } while ((interrupt & 0x01) == 0);

}
```

Wait for interrupt

- Many embedded systems are event driven – most of the processing is triggered by an event
  - In a properly designed systems events are triggered by interrupts
    - IO devices, timers, etc.
- Waiting for something requires that you have a loop that repeats until the event occurs
  - Since events are triggered by interrupts there is no need to do any checking between the interrupts
  - You can put the processor to idle/sleep state and wake up when an interrupt occurs – then check and processes events and when you are done put the processor back to sleep again
- ARM has a special instruction that puts processor to sleep until an interrupt occurs.
- PicoSDK implements a sleep function for waiting for an event (~interrupt) `best_effort_wfe_or_timeout`

Event handler

- A practical approach to event handling is to put events into a queue. Event dispatcher then reads the queue and processes them in order
- Queuing preserves event order even if the processing would be delayed



PicoSDK queue

- The simplest way to handle communication between ISR and main program is to use PicoSDK's queue
  - It is multicore and interrupt safe
  - It is part of pico_stdlib

```
#include "pico/stdlib.h"
#include "pico/util/queue.h"

static queue_t events;

static void gpio_handler(uint gpio, uint32_t event_mask)
{
    int button_value = 1;
    queue_try_add(&events, &button_value);
}


int main() {

    // Initialize chosen serial port
    stdio_init_all();

    queue_init(&events, sizeof(int), 10);

    int value = 0;
    // Loop forever
    while (true) {
        while(queue_try_remove(&events, &value)) {
            printf("Got event: %d\r\n", value);
        }
        // do something else
    }
}
```

- The code on the right shows the principle of using PicoSDK queue.
- It is not a fully functional example!
  - See GPIO and queue documentation for more information

https://www.raspberrypi.com/documentation/pico-sdk/high_level.html#queue